

Notation, Abbreviations, and Hypothesis Definitions

Mathematical Symbols and Notation

Table 1: Mathematical symbols and notation used throughout this manuscript. All scoring quantities are defined formally in §??.

Symbol	Description
N	Corpus size (total deduplicated papers)
n	Subfield paper count
T	Time span in years (for CAGR computation)
$w(a)$	Citation-weighted assertion score: $\log(1 + \text{citations}) \cdot \text{confidence}$
$\text{score}(H)$	Aggregate evidence score for hypothesis H , range $[-1, 1]$
$S(H)$	Set of supporting assertions for hypothesis H
$C(H)$	Set of contradicting assertions for hypothesis H
$A(H)$	Set of all assertions for hypothesis H
c	Assertion confidence, range $[0, 1]$
d	Assertion direction: supports, contradicts, or neutral
k	Number of latent topics
ϵ	Numerical stability constant (10^{-10})
CAGR	Compound annual growth rate
t	Publication doubling time

Abbreviations and Acronyms Used

Table 2: Abbreviations and acronyms used in this manuscript, listed alphabetically.

Abbreviation	Definition
AIF	Active Inference
API	Application Programming Interface
CAGR	Compound Annual Growth Rate
DCM	Dynamic Causal Modelling
DOI	Digital Object Identifier
EBM	Energy-Based Model
EFE	Expected Free Energy
FAIR	Findable, Accessible, Interoperable, Reusable
FEP	Free Energy Principle
FEPS	Free Energy Projective Simulation
HITS	Hyperlink-Induced Topic Search
JSON	JavaScript Object Notation
JSONL	JSON Lines (newline-delimited JSON)
LLM	Large Language Model

Standard Hypothesis Definitions and Identifiers

Table 3: Standard hypothesis definitions and identifiers tracked throughout the meta-analysis. Each hypothesis is linked to its primary domain scope within the A/B/C taxonomy.

ID	Hypothesis
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- | | |
|----|---|
| H1 | FEP Universality: The Free Energy Principle applies universally to |
| H2 | AIF Optimality: Active Inference agents achieve optimal decision-n |
| H3 | Markov Blanket Realism: Markov blankets correspond to real phys |
| H4 | Predictive Coding: Cortical hierarchies minimize prediction errors v |
| H5 | Scalability: Active Inference scales to complex, high-dimensional en |
| H6 | Clinical Utility: Active Inference provides clinically useful models of |
| H7 | Morphogenesis: The FEP explains morphogenetic and development |
| H8 | Language AIF: Active Inference provides a viable framework for lan |

Glossary of Key Terms

lp12cm Glossary of key terms used in this manuscript, including pipeline-specific concepts, statistical methods, and domain terminology.

Term Definition

[(continued)]

Term Definition

Active Inference A framework in which agents minimize expected free energy to select actions, unifying perception, learning, and decision-making under the Free Energy Principle.

Assertion A directed, confidence-scored claim linking a paper to a hypothesis (supports, contradicts, or neutral). The basic unit of evidence in the knowledge graph, representing a machine-extracted classification rather than a definitive human judgment.

Canonical ID The unique identifier assigned to each paper during deduplication, following the priority scheme: DOI > arXiv ID > Semantic Scholar ID > OpenAlex ID > title hash.

Checkpoint A JSON Lines snapshot of LLM extraction progress,